Smart Vehicle Simulation with Voice Assistant

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Course: projet dans la spécialité

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1 1. Introduction

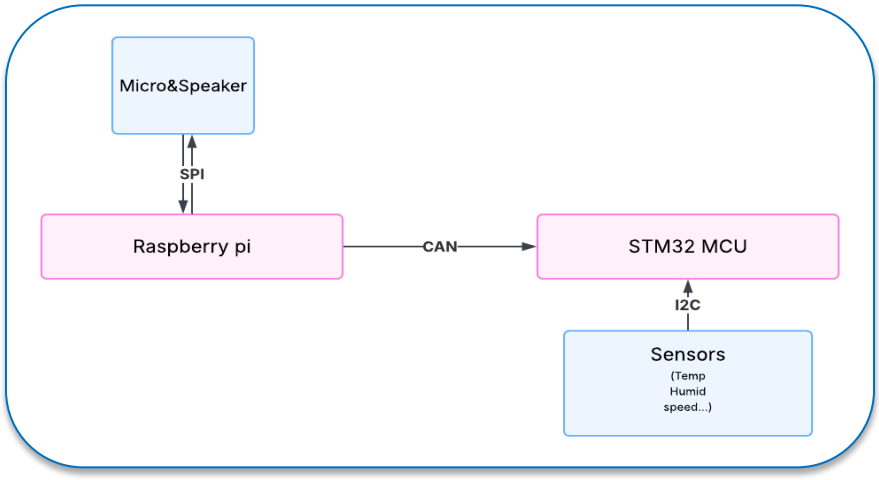
The rise of AI-powered in-vehicle assistants has transformed modern automotive design. This project provides a classroom prototype of such a system, showcasing how voice commands can control vehicle-like functions such as lights, temperature monitoring, and motor control.

2 2. Objectives

- Implement a voice assistant using Hugging Face Whisper (speech-to-text) and TTS models.
- Integrate Raspberry Pi running Yocto as the central embedded platform.
- Enable communication with STM32 via CAN bus for sensor monitoring and actuator control.
- Simulate automotive functions such as lights, humidity and temperature monitoring.

3 3. System Architecture

The following diagram illustrates the system design:



4 5. Hardware Setup

- Raspberry Pi (Model 4 or similar) running Yocto-based Linux.
- STM32 development board with CAN transceiver.
- Sensors: DS18B20 (temperature), DC motor, LDR (light), BME280 (weather).

- Microphone and speakers for voice interaction.

5 6. Software Setup

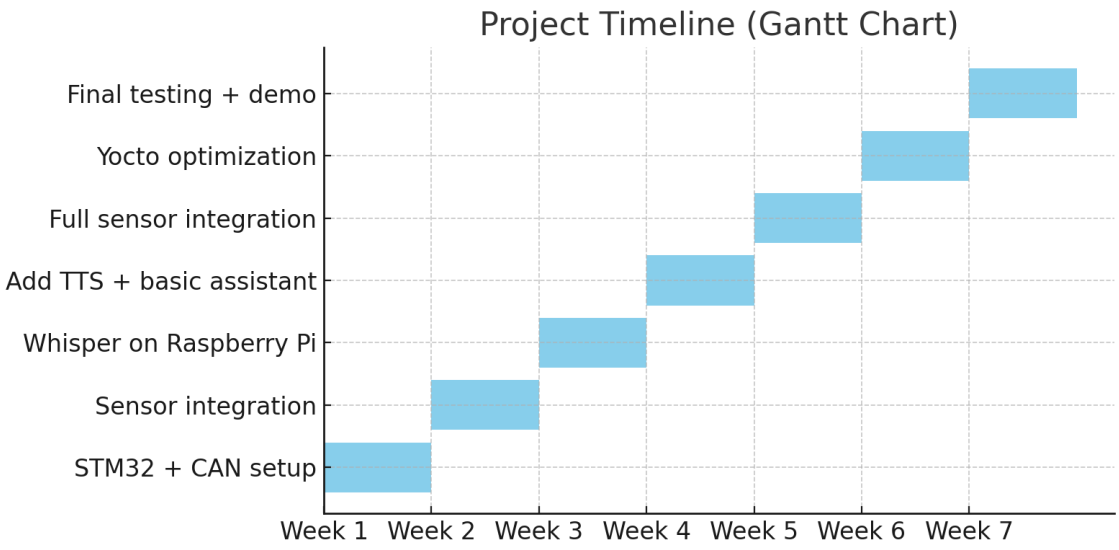
- Yocto layers: meta-raspberrypi, meta-python, meta-openembedded, meta-ivi.
- Python libraries: transformers (Whisper), pyttsx3 (TTS), python-can, sounddevice.
- STM32 firmware with CAN bus communication configured in STM32CubeIDE.

6 7. Workflow

1. User speaks into microphone connected to Raspberry Pi.
2. Whisper converts speech to text.
3. Python logic interprets the command.
4. Raspberry Pi sends CAN message to STM32.
5. STM32 executes action (light on/off, motor control, sensor readout).
6. Response sent back → Raspberry Pi generates spoken reply via TTS.

7 8. Project Timeline

The project timeline is illustrated in the following Gantt-style chart.



8 10. Expected Results

The system will act as an embedded automotive assistant running on Raspberry Pi with Yocto. It will process natural voice commands, control STM32 actuators over CAN, and provide audio responses. The classroom demo will simulate modern vehicle functions and highlight the integration of AI, IoT, and embedded systems.